Can Identity Verification Improve Fertilizer Targeting in Nigeria?

Nneka Osadolor and Abiola Oyebanjo*

Context

Kaduna State, in north-western Nigeria, is the country's third most populous state. Agriculture is its main livelihood. Close to 80 percent of households are involved in farming (KDBS, 2017). The Kaduna State Agricultural Development Agency (KADA) runs the state's fertilizer subsidy and keeps its farmer registry. The Kaduna State Residents Identity Management Agency (KADRIMA) keeps the state's register of residents. Since 2018, the state has issued a biometric Residents Card to adults living in Kaduna. The card is tied to the National Identification Number (NIN), issued by the National Identity Management Commission (NIMC). Since 2021, residents have needed the card or a NIN to use state services. In December 2025, a pilot randomized card distribution among 13,523 residents to test distribution methods and the card's effect on access to social protection and financial services (Cohen et al., 2025). Records from the pilot sample have been matched against the beneficiary rosters of several state agencies, including KADA.

Work led by the 1864 Research and Innovation Institute is in progress to link the two agencies' records through the NIN. The linkage verifies KADA's beneficiary list against KADRIMA's register. It changes what is known about each entry when the list is fixed. It does not change eligibility rules, the subsidy amount, or how fertilizer is distributed.

Evaluation Design

Subsidised fertilizer does not reliably reach eligible farming households (Jayne and Rashid, 2013; Jayne et al., 2018). The farmer registry shows who is registered as a farmer, but it cannot tell whether two entries belong to the same household or whether an identity is genuine. On the other hand, the identity register shows who each person is, where they live, and which household they belong to, but holds no farming data.

As a result, fertilizer leaks in two ways (Mason and Jayne, 2013). On the beneficiary side, several members of one household register and collect fertilizer separately, people collect using a relative's identity, and traders or civil servants register as farmers and resell the fertilizer. On the administrative side, names are added at the request of local leaders, and leftover bags go to relatives or whoever is present (Pan and Christiaensen, 2012; Kilic et al., 2015; Basurto et al., 2020).

Linking the two records allows the farmer registry to be checked against verified identities and households before distribution (Christen, 2012). Repeat households and invalid identities are removed, and entries whose address falls outside the subdivision are flagged for review. Officers distribute only to people on the list, so these removals should reduce leakage directly (Muralidharan et al., 2016; Banerjee et al., 2020).

Some channels remain. A person can still collect using a relative's identity when that relative is legitimately on the list. Because the identity register holds no farming data, non-farmers with valid identities stay on the list, and farmers who were never registered are not added (compare Aiken et al., 2022). We therefore expect the verified list to reduce leakage and improve targeting, and the trial measures by how much.

*1864 Research and Innovation Institute, 1864institute.org.

The study asks two questions.

- Does a verified beneficiary list improve the targeting of fertilizer?
- Who gains fertilizer as a result, and how does this affect their farming outcomes (fertilizer use, crop yields, household income, and food security)?

To answer them, KADA and KADRIMA will work with the research team on a randomized controlled trial across 53 subdivisions of Kaduna State. Randomization happens at two levels. First, subdivisions are randomly assigned to receive the verified list (treatment) or not (control). Second, within each subdivision, survey households are drawn at random, both from KADA's list and from a fresh listing of farming households in randomly selected villages. In treatment subdivisions, KADRIMA, working with the research team, will match KADA's pre-season list against its register before distribution, and KADA will use the verified list as the distribution list. Farmers without a national identity record will be kept on the list so that verification does not exclude them (Muralidharan et al., 2025). Control subdivisions will distribute from KADA's original list and will receive a verified list the following season. Because fertilizer can be reallocated during the season, the research team will record each subdivision's allocation before and after cleaning, and any bags moved between subdivisions.

In both arms, KADA and the research team will jointly hold the distribution list and log any change to it, KADA officers will record each recipient's identifier at collection, and independent monitors will make unannounced visits to centres. The study therefore estimates the effect of a verified list when distribution is already closely monitored (Olken, 2007).

Targeting is measured in two ways, identically in both arms.

- **Leakage.** A recipient counts as leakage if they belong to a household that has already collected, use an invalid or borrowed identity, or live outside the subdivision. Gaps between bags received and bags issued are also measured. Collection using another person's identity is detected when a listed farmer reports in the survey that they did not collect, but the distribution record shows a collection under their identity. After distribution, KADRIMA and the research team will also match every recipient in all 53 subdivisions against KADRIMA's register, using the same criteria in both arms. This also shows whether people removed from a treatment list collected in a neighbouring control area.
- **Inclusion.** The household survey asks whether eligible farmers received fertilizer. Because its sample includes the fresh village listing, it covers farmers who were never registered. Effects on leakage, fertilizer received, and farming outcomes are reported separately for women and persons with disabilities.

Additional measures are changes made to the list after it was fixed, and cross-checks of distribution records against monitors' observations. The same survey measures fertilizer use, crop yields, household income, and food security. The main estimate will be the effect of assigning a subdivision a verified list.

Data protection is built in. All matching takes place within KADRIMA, carried out by KADRIMA staff working with the research team. Only subdivision-level counts leave the agency, and every research use of the data is specified in a Memorandum of Understanding.

The study will show whether privacy-preserving data sharing between government agencies can reduce leakage in a large public subsidy program, and whether that reaches farmers' fields.

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